

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 8 — Magnetism

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

Q.1: C — Electrons 1

Explanation: According to **domain theory**, magnetism arises from the motion of **charged particles**. Electrons orbit the nucleus and also spin, each producing a tiny magnetic field. Protons and neutrons are in the nucleus; their magnetic contribution is negligibly small compared to electrons.

Q.2: C — 10^{12} 1

Explanation: The textbook states that a **magnetic domain** is a group of roughly 10^{12} atoms of size a few millimetres. Students commonly confuse 10^6 (a million) with 10^{12} (a trillion); the chapter specifies 10^{12} explicitly.

Q.3: C — North pole 1

Explanation: By convention, **magnetic field lines originate from the N-pole** and terminate at the S-pole outside the magnet, and travel from S to N inside it. Students often confuse the convention with the physical source of the field.

Q.4: B — Tesla 1

Explanation: The **tesla (T)** is the SI unit of magnetic field strength B, equal to $\text{N A}^{-1} \text{ m}^{-1}$. Weber is the unit of magnetic flux; Ampere is the unit of current; Henry is the unit of inductance.

Q.5: B — Induced magnet 1

Explanation: The textbook defines an **induced magnet** as “a material that becomes a magnet when placed in a magnetic field.” A permanent magnet retains magnetism without a field; diamagnetic materials are actually weakly repelled by a magnetic field.

Q.6: D — Solenoid method 1

Explanation: The chapter explicitly states that the **solenoid method** is “the most common method used for the magnetization of metals.” Domains align with the external field produced by the current-carrying coil wrapped around the metal.

Q.7: C — Soft iron 1

Explanation: **Soft magnetic materials** (like soft iron) easily magnetize and demagnetize, making them ideal for electromagnets where the field must be switched on and off. **Hard magnetic materials** (like steel) resist demagnetization and are used for permanent magnets.

Q.8: C — Zero 1

Explanation: In **diamagnetic materials** the spin and orbital motions of electrons cancel each other, giving a net magnetic field of zero per atom ($B = 0$). Paramagnetic materials have a small non-zero net field; ferromagnetic materials have a large net field.

Q.9: B — 11.3° 1

Explanation: The textbook states that Earth’s **magnetic poles are inclined at 11.3°** to the geographical poles. This is why a compass needle does not point exactly toward geographic north but deviates slightly toward the magnetic pole.

Q.10: D — Nickel 1

Explanation: **Ferromagnetic materials** include iron, cobalt, nickel, and some metallic alloys. Silver and copper are diamagnetic; aluminium is paramagnetic. Nickel is specifically listed in the chapter as a ferromagnetic material.

Q.11: C — Two complete magnets 1

Explanation: Magnets are always **bipolar (dipole)**. Cutting a bar magnet creates two smaller magnets each with its own N and S pole. A **monopole** (a magnet with only one pole) does not exist. Students often think cutting makes them unmagnetized or produces monopoles.

Q.12: D — Motion of ions in the core 1

Explanation: Earth's core mainly consists of molten iron. **Electrical currents** flowing in this slowly moving molten iron generate Earth's magnetic field; this is called the **dynamo effect**. Rotation and spinning of Earth are related to the geographical axis, not the magnetic field source.

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)**2(i) Domain Theory** 3

Domain theory of magnetism: Magnetism arises because electrons (charged particles) move around the nucleus, producing tiny magnetic fields. In a sizeable piece of material, a group of roughly 10^{12} atoms whose N-poles point in the same direction forms a **magnetic domain** (size: a few mm). 1

In an **unmagnetized material** domains are randomly oriented so their fields cancel and the net field is zero. 1

In a **magnetized material** (placed in an external magnetic field) all domains align in the direction of the applied field, giving the material a net non-zero magnetic field. 1

OR

Electrons and atomic magnetism 3

Every electron has two motions: (a) **orbital motion** around the nucleus, producing an orbital magnetic moment, and (b) **spin motion**, producing a spin magnetic moment. Each electron therefore acts as a tiny magnet. 1

In some atoms these electron magnetic moments add up to a non-zero net field, making the material magnetic; in others they cancel. 1

The **spinning nucleus** also produces a tiny magnetism, but it is negligibly small compared to electron contributions and is therefore ignored in the study of magnetism. 1

2(ii) Magnetic Field 3

Magnetic field: "The region or space around a magnet where it exerts a force on other magnetic poles is called the magnetic field." 1

SI unit of B: Tesla (T), which equals $\text{N A}^{-1} \text{m}^{-1}$. 1

Direction of B at a point: determined by placing a north-pole at that point. The force experienced by the north-pole gives the direction of the magnetic field at that point. On a curved field line, the direction is found by drawing a tangent at that point. 1

OR

Magnetic field lines 3

Magnetic field lines are curves that represent the magnetic field graphically. Three properties:

1 per property

- They originate from the **N-pole** and terminate at the **S-pole** outside the magnet (and travel S to N inside).
- They **never intersect** each other (because the field can have only one direction at any given point).
- Where field lines are **closer together** the field is stronger; where they are farther apart the field is weaker.

2(iii) Relative Strength of Magnetic Field 3

Magnetic field lines give **both direction and strength** of the field at any point. 1

The field is **stronger where lines are closer together** and weaker where lines are farther apart. This spacing is called the **relative strength (degree of closeness)** of the field lines. 1

By placing two like poles close together a **neutral zone** (field-free region) forms between them where the field lines repel each other and cancel. Placing unlike poles close together strengthens the field between them. 1

OR

Magnetic Shielding 3

Magnetic shielding: the phenomenon in which some region of space is made free from a magnetic field by suitably arranging magnets to create a neutral zone. 1

The **neutral zone** is created by placing two N-poles side by side so their field lines repel each other, making a field-free region between them. Devices like power inverters and magnetic sensors must be shielded from external fields. 1

Three shield materials: **iron, nickel, and cobalt**. Shields are made with **rounded corners** because magnetic field lines cannot easily turn at a 90° angle, so sharp corners would let field lines pass through.

1

2(iv) Induced Magnetism & Stroking Method 3

Induced magnetism: the phenomenon in which a non-magnetic material is made into a magnet using various techniques. A material that becomes a magnet when placed in a field is called an **induced magnet**.

1

Stroking method: a permanent magnet is stroked from one end of a steel bar to the other, always in the same direction. The poles of the bar material align with the stroking magnet. 1

After one stroke the magnet is lifted and returned to the starting end before the next stroke. The end where stroking **finishes** becomes a **south pole** (opposite to the pole of the magnet that last left that end). 1

OR

Hammering Method 3

In the **hammering method** a bar of metal is placed inside a strong external magnetic field and then **hammered gently**. The vibration caused by hammering allows domains to align in the direction of the applied field. 1

This method is mainly used for the magnetization of **steel**. 1

The magnetization can be **increased by heating** the metal bar slightly before hammering, which makes domain alignment easier. 1

2(v) Heating Method (Demagnetization) 3

The **heating method** is usually used to **demagnetize** a material. Heat speeds up the random movement of already-aligned domains, causing **misalignment** of domains, so the material loses its magnetization. 1

Demagnetization by heat is irreversible once the material cools randomly. 1

The **magnetic Seebeck effect (thermo-magnetism)**: recent evidence shows that heating can also generate a magnetic field in certain materials. This is the reverse of the usual demagnetization effect and is still an active research topic. 1

OR

Solenoid Method & AC vs DC 3

A **solenoid** is a coil of wire wrapped around a cylindrical core. When a metal is placed inside and current flows through the coil, the magnetic field of the solenoid acts as an external field and aligns the domains of the metal, magnetizing it. 1

Direct current (DC) keeps the polarity of the solenoid constant, so domains align in one fixed direction; the material is magnetized. 1

Alternating current (AC) reverses the polarity every half cycle: in the first half cycle the material magnetizes in one direction, then demagnetizes; in the second half it magnetizes in the opposite direction, then demagnetizes again. The net result is **demagnetization**. 1

2(vi) $B = \mu_0 n I$ (Solenoid Formula) 3

The magnetic field strength inside a solenoid is given by: $B = \mu_0 n I$ 1

- **B**: magnetic field strength, unit = tesla (T)
- μ_0 : permeability of free space (vacuum) = $4\pi \times 10^{-7} \text{ N A}^{-2}$
- **n**: number of turns per unit length (m^{-1}), $n = N/L$
- **I**: current through the solenoid, unit = ampere (A)

1 for correct definitions and units.

Permeability is the property of a material measuring how much it permits a magnetic field to pass through it. 1

OR

Solenoid Calculation 3

Given: $N = 200$, $L = 50 \text{ cm} = 0.5 \text{ m}$, $I = 2 \text{ A}$, $\mu_0 = 4\pi \times 10^{-7} \text{ N A}^{-2}$

Step 1: $n = N/L = 200/0.5 = 400 \text{ m}^{-1}$ 1

Step 2: $B = \mu_0 n I = (4\pi \times 10^{-7})(400)(2)$ 1

$B = 4\pi \times 10^{-7} \times 800 = 3200\pi \times 10^{-7} \approx 1.005 \times 10^{-3} \text{ T}$ 1

2(vii) Temporary vs Permanent Magnets 3

Temporary Magnet	Permanent Magnet
Behaves as a magnet only when current flows (presence of field)	Behaves as a magnet even without current or external field
Also called electromagnet; made of soft magnetic materials	Made of hard magnetic materials
Examples: iron nails, screws, metal bolts, kitchen utensils	Examples: iron ore (Alnico), cobalt, nickel

1 per correct row (any two rows for full marks plus examples).

OR

Soft iron vs Steel 3

Soft iron is a soft magnetic material: it easily magnetizes and just as easily demagnetizes when the current is removed. This is essential for electromagnets, which must turn on and off repeatedly. 1

Steel is a hard magnetic material: once magnetized it retains its magnetism even when the external field is removed. This property makes it ideal for permanent magnets. 1

In general: **soft magnetic materials** (low retentivity) are used for electromagnets/temporary purposes; **hard magnetic materials** (high retentivity) are used for permanent magnets. 1

2(viii) Diamagnetic vs Paramagnetic 3

Diamagnetic Materials	Paramagnetic Materials
Net magnetic field per atom = zero (spin and orbital motions cancel)	Net magnetic field per atom = small but non-zero
Slightly repelled by an external magnetic field	Feebly attracted to an external field; align in field direction
Do not retain magnetism after field is removed	Do not retain magnetism after field is removed
Examples: copper, zinc, bismuth, silver, gold, glass, water	Examples: tungsten, aluminium, lithium, sodium

1 net field, 1 behaviour, 1 examples.

OR

Ferromagnetic Materials 3

Ferromagnetic materials: materials that have a net magnetism at the atomic level even without an external field. 1

Three distinguishing properties: (a) **strongly attracted** to a magnet (unlike paramagnetic which are only feebly attracted), (b) they **retain magnetism** after the external field is removed, and (c) their magnetic field intensity is **very large** (unlike the small field of paramagnetic materials). 1

Examples: **iron, cobalt, nickel**, and some metallic alloys. 1

2(ix) Magnetic vs Non-Magnetic Materials 3

Magnetic Materials	Non-Magnetic Materials
Attracted to a magnet	Not attracted to a magnet
Atomic states are aligned	Atomic states are random
Respond to a magnetic field	Do not respond to a magnetic field

1 per correct difference (3 marks for three differences).

Example of magnetic material: **nickel**. Example of non-magnetic material: **rubber (or plastic/wood/copper)**.

OR

Can field lines intersect? 3

No, magnetic field lines can never intersect each other. 1

The direction of the magnetic field at any point is determined by the direction of the force on a north-pole placed at that point. 1

If two field lines intersected at a point, then the field at that point would have **two directions simultaneously**, which is physically impossible. Therefore field lines never cross. 1

2(x) Earth's Magnetic Field and the Dynamo Effect 3

Earth is a large magnet with its own magnetic field called the **geomagnetic field**. It extends millions of kilometres into space but has a very small field strength. 1

Earth's core mainly consists of **molten iron**. Electrical currents flowing in this slowly moving molten iron generate the magnetic field of Earth; this is called the **dynamo effect**. 1

Earth's field acts as a protective shield, saving life from **harmful cosmic rays** and charged particles and radiation from the Sun. The field resembles that of a bar magnet. 1

OR

Geographical vs Magnetic Poles 3

Earth spins about its **geographical axis**, with the geographical north pole at the Arctic and the south pole at the Antarctic. 1

The **magnetic poles** of Earth are inclined at an angle of **11.3°** to the geographical poles. They are therefore not at the same location. 1

Practical effect: a compass needle **does not point exactly toward geographic north**; instead it points 11.3° away toward the magnetic pole. Navigators must account for this **magnetic declination**. 1

2(xi) Speakers 3

A speaker uses two magnets: a **permanent magnet** of strong magnetization fixed at the centre of the cone (diaphragm) and an **electromagnet (voice coil)** attached to the centre of the permanent magnet. 1

When a sound signal (as alternating electric current) flows through the electromagnet coil, it produces an **alternating magnetic field** which interacts with the permanent magnet's field, causing the electromagnet to move to and fro. Since the cone is attached to the electromagnet, it vibrates and produces sound. 1

The commonly used permanent magnet material in speakers is **neodymium**. A larger permanent magnet produces louder sound. 1

OR

Magnetic Recording 3

Magnetic recording is the method of saving sounds, pictures, and data in the form of electrical signals by **selective magnetization** of portions of a magnetic material. 1

A **tape head** moves across the tape. The signal current flowing through the tape head creates a magnetic field that aligns domains in the tape's magnetic emulsion, recording the data. Reading is the reverse process. 1

Three recording media: **iron oxide, cobalt, chromium oxide** (and pure iron). Two recording devices: **magnetic tape recorders and disk recorders**. 1

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Domain Theory of Magnetism 5

(a) How electrons produce magnetism at the atomic level 1

Everything is made of atoms with a nucleus (positive charge) and orbiting electrons (negative charge). In late 20th century science established that the basis of magnetism is the **motion of charged particles**. Each electron has two motions: (i) orbital motion around the nucleus producing an **orbital magnetic moment**, and (ii) spin producing a **spin magnetic moment**. In some atoms these moments add up to a non-zero net field, making the material magnetic.

(b) Definition and typical size of a domain 1

A **magnetic domain** is a group of atoms in a material whose N-poles all point in the same direction. A typical domain contains roughly **10^{12} atoms** and has a size of a few millimetres.

(c) Domain arrangement before and after magnetization 2

- **Before (unmagnetized):** domains are randomly oriented; their fields cancel and the net magnetic field is zero.
- **After (magnetized):** domains align in the direction of the applied external magnetic field; the material has a net non-zero magnetic field and acts as a magnet.

(d) Effect of heavy heating 1

Heavy heating speeds up random movement of already-aligned domains, causing them to **misalign**. The material loses its magnetization. This is why the heating method is used for demagnetization.

OR

Magnetic Field and Field Lines 5

(a) Plotting field lines 1

Using a **compass**: place the bar magnet on a sheet; note where compass needles point around it; the pattern of arrows traces the field lines. Using **iron filings**: scatter iron filings around the magnet; they align along field lines showing the same pattern as compass needles.

(b) Direction of field lines 1

Outside the magnet: lines emerge from the **N-pole** and enter the **S-pole**. Inside the magnet body: lines travel from **S-pole to N-pole**, forming closed loops.

(c) Spacing and field strength 1

The **closer the field lines**, the stronger the magnetic field at that point. Conversely, widely spaced lines indicate a weaker field. The relative strength is the degree of closeness of the field lines.

(d) Neutral zone 2

A **neutral zone** (field-free region) is formed by placing two N-poles side by side. Their field lines seem to repel each other, creating a region between them where the net field is zero. This phenomenon is called **magnetic shielding of a region** and is exploited to protect sensitive electronics from stray fields.

3(ii) Methods of Induced Magnetization + Solenoid Calculation 5

Three methods of magnetization: 1

STROKING METHOD

Stroke permanent magnet along bar (one direction)
--> Domains align with stroking field
--> End where stroking ENDS = south pole
[Used for: steel bars; easy; moderate permanence]

HAMMERING METHOD

Place bar in strong external field
--> Hammer gently (domains vibrate into alignment)
--> Heat slightly before hammering to improve alignment
[Mainly for: steel magnetization]

SOLENOID METHOD (most common)

Wrap insulated copper wire on metal rod
--> Pass DC current through coil
--> Solenoid field aligns domains of metal inside
[DC = magnetizes; AC = demagnetizes]

Formula and definitions: 1

$B = \mu_0 n I$ where B = field strength (T), $\mu_0 = 4\pi \times 10^{-7} \text{ N A}^{-2}$, $n = N/L$ (turns per unit length, m^{-1}), I = current (A).

Calculation: 2

Given: $N = 100$, $L = 25 \text{ cm} = 0.25 \text{ m}$, $I = 3 \text{ A}$

$$n = N/L = 100/0.25 = 400 \text{ m}^{-1}$$

$$B = \mu_0 n I = (4\pi \times 10^{-7})(400)(3) = 4800\pi \times 10^{-7}$$

$$B \approx 1.508 \times 10^{-3} \text{ T} \approx 1.51 \times 10^{-3} \text{ T}$$

Most commonly used: 1 The **solenoid method** is most commonly used because it can be applied to any shape of metal, the field strength is easily controlled by adjusting current or number of turns, and it can rapidly magnetize large quantities of material.

OR

Permanent Magnets vs Electromagnets 5

Feature	Permanent Magnet	Electromagnet
Material	Hard magnetic materials (steel, Alnico, cobalt)	Soft magnetic materials (soft iron)
How magnetism is produced	Magnetized once; retains magnetism permanently	Magnetic field produced by electric current
Strength variable?	No; magnetism does not vary in strength	Yes; strength varied by changing current or turns
Poles changeable?	No; poles cannot be altered	Yes; poles can be changed by reversing current

2 for comparison table.

Applications of permanent magnets: induction cookers, MRI machines, particle accelerators, transformers, automotive industry, aerospace, medical devices, semiconductor manufacturing. (any 4) 1

Applications of electromagnets: electric fans, electric motors, door bells, hair dryers, electric razors, hair trimmers, MRI scan coils, communication devices, power circuits. (any 4) 2

3(iii) Three Types of Magnetic Materials 5

Feature	Diamagnetic	Paramagnetic	Ferromagnetic
Net field per atom	Zero ($B = 0$)	Small, non-zero ($B \neq 0$)	Large, non-zero
Behaviour in external field	Slightly repelled; field intensity small and opposite to applied field	Feebly attracted; align in direction of applied field; feeble magnetization	Strongly attracted; align parallel to applied field; strongly magnetized
Magnetism retained after field removed?	No	No	Yes
Examples	Copper, zinc, bismuth, silver, gold, glass, water, marble	Tungsten, aluminium, lithium, sodium	Iron, cobalt, nickel, metallic alloys

2 for complete table. 1 per correct type description (3 marks for all three).

The chapter calls **diamagnetic** materials “non-magnetic materials” because they have zero net magnetic field in the absence of an external field ($B = 0$). 1

OR

Magnetic vs Non-Magnetic + Shielding 5

Magnetic Materials	Non-Magnetic Materials
Attracted to a magnet	Not attracted to a magnet
Atomic states are aligned	Atomic states are random
Respond to a magnetic field	Do not respond to a magnetic field
Have a magnetic field around them	Do not have a magnetic field around them
Can attract and repel other magnetic materials; examples: Nickel, Cobalt, Steel, Iron	Cannot attract or repel magnetic material; examples: Rubber, Plastic, Wood, Copper

2 for five-point table.

Magnetic Shielding: 1 the phenomenon in which a region of space is made free from a magnetic field. It is needed because stray fields can alter the functioning of sensitive devices like power inverters and magnetic sensors.

The **neutral zone** is the field-free region created between two N-poles placed side by side; their opposing field lines cancel there. 1

Two suitable shield materials: **iron and nickel** (also cobalt). Shields are made with **rounded corners** because magnetic field lines cannot easily turn at 90° ; rounded edges allow the field to flow through the shield material rather than penetrating the interior. 1

3(iv) Earth's Magnetic Field 5

(a) Dynamo effect 1

Earth's core mainly consists of **molten iron**. Electrical currents flowing in the slowly moving molten iron generate Earth's magnetic field; this process is called the **dynamo effect**. Earth is therefore effectively a large magnet with its own magnetic field, also known as the **geomagnetic field**.

(b) Resemblance to a bar magnet 1

Earth's magnetic field resembles the field of a bar magnet: it has a N-pole and S-pole, and field lines emerge at one pole, loop around, and enter at the other. On the Sun-facing side the field is compressed by solar winds; on the opposite side it extends several million kilometres into space.

(c) Angle between geographical and magnetic poles 1

Earth's **magnetic poles are inclined at 11.3°** to the geographical poles. The magnetic axis and geographical axis are therefore not the same, which is why a compass needle does not point exactly toward geographic north but instead deviates 11.3° toward the magnetic pole.

(d) Protection from cosmic rays and solar radiation 1

Earth's magnetic field acts as a **protective shield** around Earth. It deflects harmful **cosmic rays** coming from outer space and the **charged particles and radiation** (solar wind) streaming from the Sun. Without this shield, life on Earth could not survive the constant bombardment of high-energy particles.

(e) Aurora 1

The **Aurora** (dancing lights in the sky) forms at the northern and southern poles of Earth. Earth's magnetic field has **polar cusps** where charged particles from the Sun enter the atmosphere. When these particles collide with atoms of air (gases), they **ionize** them, which causes emission of light of different colours depending on the nature of the gas, producing the Aurora.

OR

Magnetic Recording + Speakers and Microphones 5

(a) Magnetic Recording 2

Magnetic recording is the method of saving sounds, pictures, and data as electrical signals by **selective magnetization** of portions of a magnetic material.

WRITING (Recording):

- Signal current --> Tape head (electromagnet)
- > Tape head moves over magnetic tape
- > Magnetic field of tape head aligns domains in tape emulsion
- > Pattern of domains = encoded data

READING (Playback):

- Tape moves under tape head
- > Aligned domains in tape induce a current in tape head
- > Current decoded as audio/video/data signal

Three recording media: **iron oxide, cobalt, chromium oxide**. Two recording devices: **magnetic tape recorders** (audio/video) and **disk recorders** (computer storage). Also: magnetic drums, core and bubble units. 1

(b) Speakers and Microphones 2

Speaker: A permanent magnet is fixed at the centre of the cone (diaphragm). An electromagnet (voice coil) is attached at the same centre. When an audio signal (AC) flows through the voice coil, its alternating field interacts with the permanent magnet's field, pushing and pulling the voice coil back and forth. The attached cone vibrates, moving air and producing sound. Permanent magnet material: **neodymium**.

Microphone: works in reverse. Sound waves cause the cone (diaphragm) to vibrate, which moves the electromagnet within the field of the permanent magnet. This movement generates an **alternating electric current** in the coil, which becomes the audio signal. The microphone therefore converts mechanical energy (vibration) into electrical energy, while the speaker converts electrical energy into mechanical energy (sound). 1